

Combinational Assignment

1. Implement the following functions using

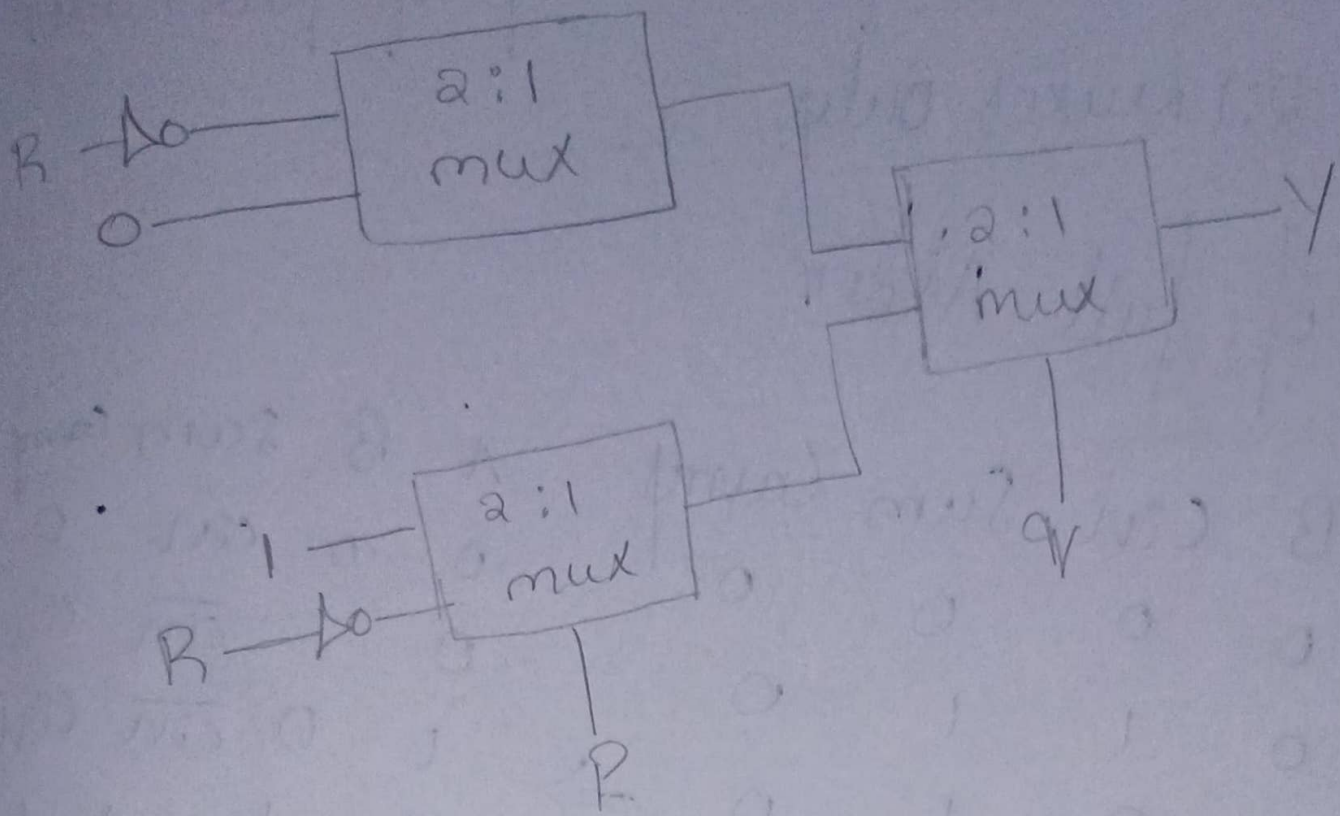
i. 2:1 mux and not gates.

ii. 4:1 mux

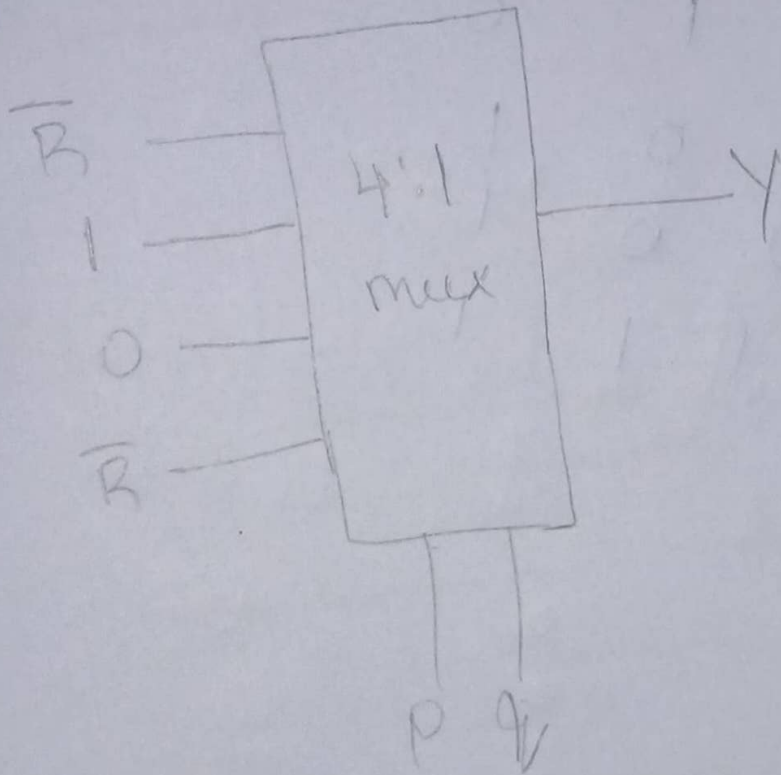
a. $F(A, B, C) = \bar{A}C + B\bar{C}$
 $= \bar{A}C(B + \bar{B}) + (A + \bar{A})B\bar{C}$
 $= \bar{A}BC + \bar{A}\bar{B}C + AB\bar{C} + \bar{A}B\bar{C}$
 $= m(1, 2, 3, 6)$

TT (using 2:1 mux and not gates)

A	B	C	Y (output)	A	B	Y
0	0	0	0	0	0	C
0	0	1	1	0	1	$\bar{C}$
0	1	0	1	1	0	0
0	1	1	1	1	1	$\bar{C}$
1	0	0	0			
1	0	1	0			
1	1	0	1			
1	1	1	0			



ii) using 4:1 mux



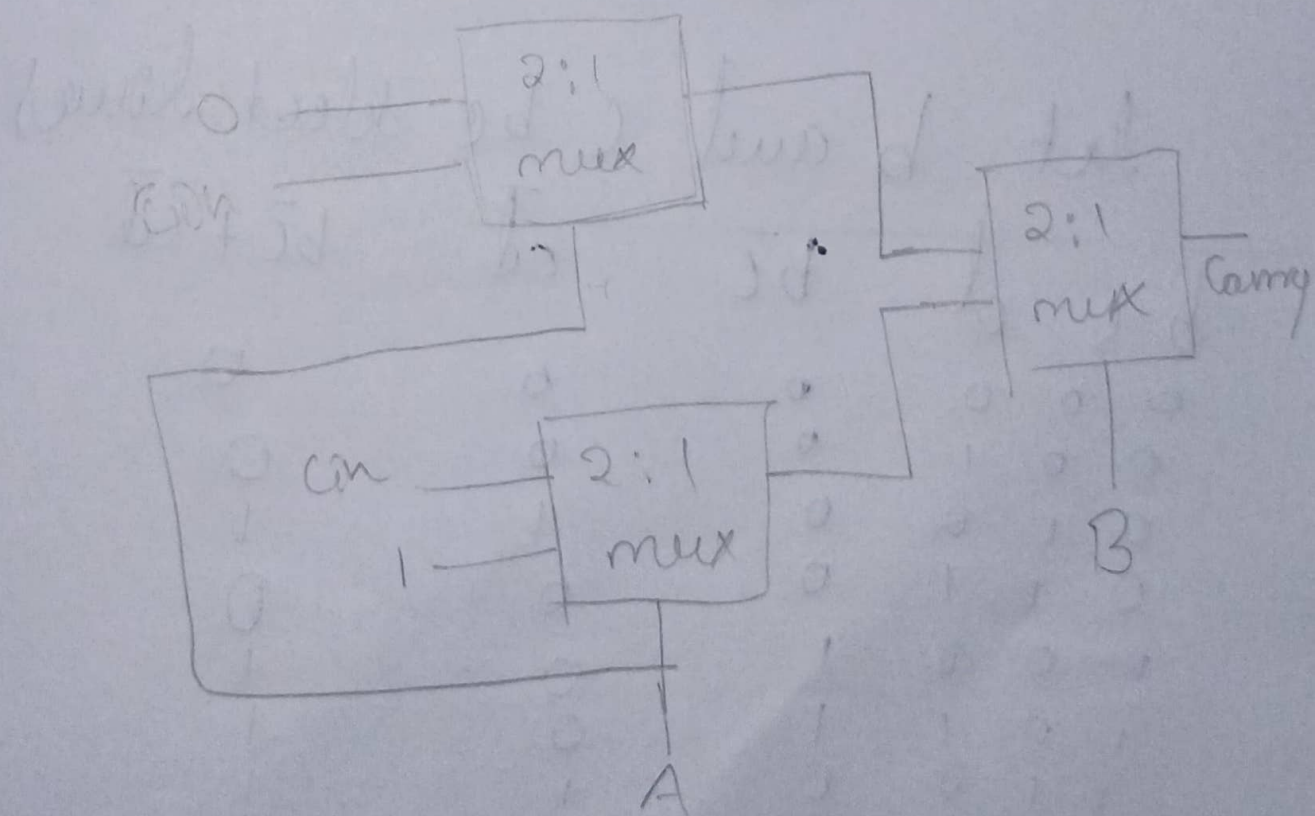
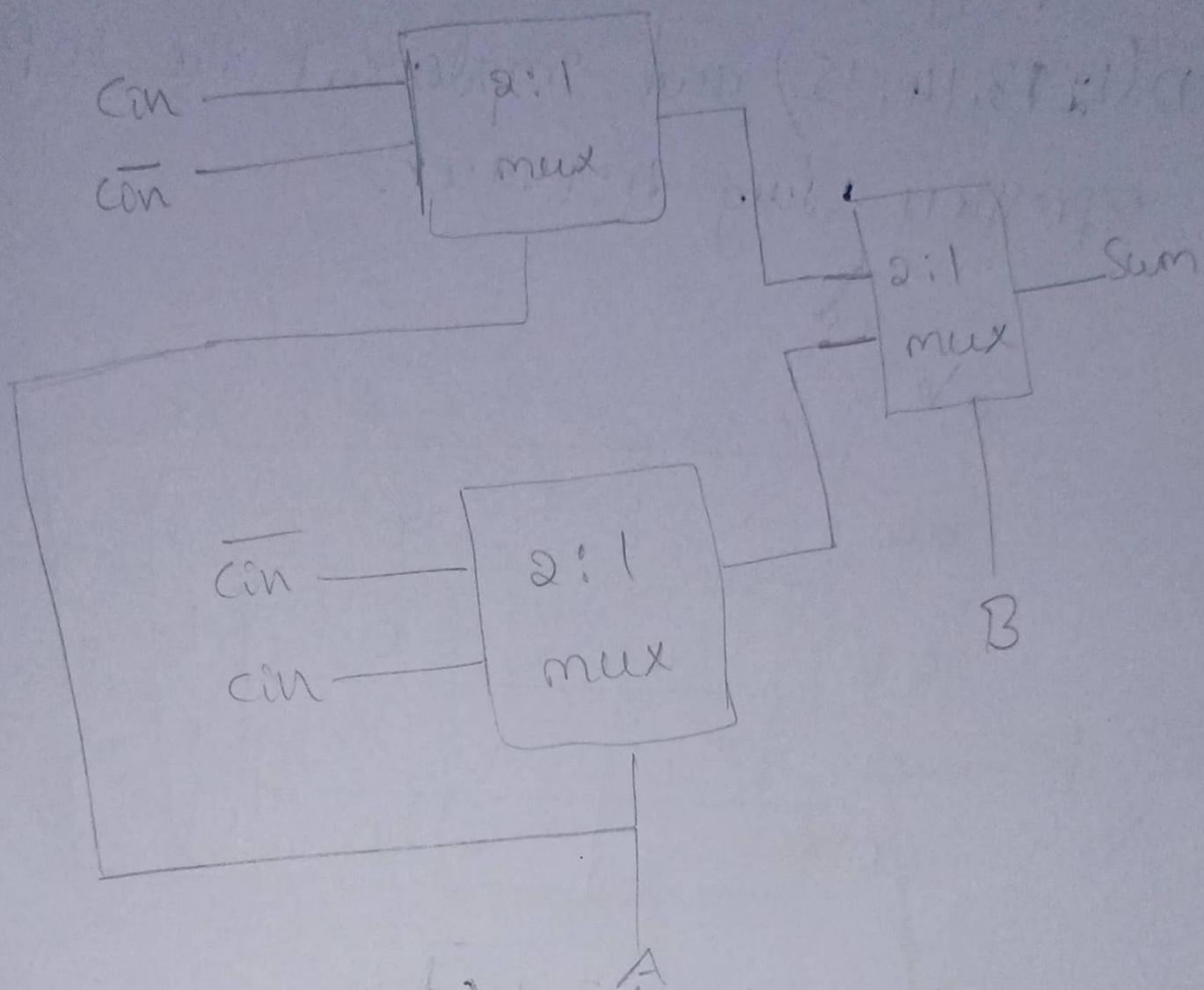
2. Implement the functionality of full Adder using 2:1 muxes only.

• TT of full Adder

A	B	Carry	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

A	B	Sum	Carry
0	0	0	0
0	1	$\overline{0}$	0
1	0	$\overline{0}$	0
1	1	0	1

$\Rightarrow$



3) Simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + d(12, 13, 14, 15)$ and implement the SOP expression using 4:1 Mux.

ab \ cd					
		00	01	11	10
00	0	0	1	0	1
01	1	4	1	5	0
11	X	12	X	13	X
10		8	9	0	11

$$Y = b\bar{c} + c\bar{d}$$

Let b and c be select lines

b	c	d	$b\bar{c}$	$c\bar{d}$	$b\bar{c} + c\bar{d}$
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	0

b	c	$y_2 = b\bar{c} + c\bar{d}$
0	0	0
0	1	$\bar{d}$
1	0	1
1	1	$\bar{d}$

